

Hotel Booking Analytics

Hospitality Booking Data to Improve Business Performance

Capstone Project
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Tools used:

- Python (Pandas, Matplotlib, Seaborn)
- Power BI
- Excel

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Project Background

Business Context

The Hotel Booking Analytics dataset captures reservation activity across hotels, customers, market segments, and booking outcomes. The dataset is designed to support hospitality analytics, revenue management, occupancy analysis, customer behavior analysis, and reservation performance monitoring. By combining booking transactions with hotel, customer, market segment, and reservation status information, analysts can evaluate booking patterns, cancellation behavior, guest preferences, distribution channel effectiveness, and overall hotel performance.

Project Goal

To transform historical hotel booking data into actionable business insights that support strategic decision-making and improve overall hotel performance.



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Business Problem & Objectives

Business Problem

How can hospitality booking data be leveraged to identify the key factors influencing booking cancellations, occupancy patterns, customer value, distribution channel effectiveness, and reservation outcomes in order to support data-driven decision-making and improve hotel performance?

Research Objectives

1. Identify factors influencing booking cancellations.
2. Analyze hotel occupancy patterns.
3. Evaluate customer segment performance.
4. Assess distribution channel effectiveness.
5. Examine reservation outcomes.



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Dataset Overview

Dataset Used for Analysis

Total five dataset .csv files

- Primary transactional table contains detailed reservation information for hotel guests.
- Data consists of historical arrivals: July 2015–August 2017

Filename	Description	Records
Bookings.csv	Primary transactional table containing detailed reservation information for hotel guests.	119,390

- Four lookup tables

Filename	Description	Records
Hotels.csv	Hotel master reference table and contains information about the hotels included in the dataset.	2
Customer Types.csv	Customer Types table contains customer classification information, deposit policies, and meal package selections.	36
Market Segments.csv	Market Segments table contains booking acquisition information and identifies how reservations were generated.	26
Reservation Status.csv	Reservation Status table contains reservation outcome classifications used throughout the dataset.	3

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Data Dictionary

Bookings table

Column Name	Description	Unit
Booking_ID	Unique identifier assigned to each booking record.	Identifier
Hotel_ID	Identifier of the hotel associated with the booking.	Identifier
Market_Segment_ID	Identifier of the booking market segment and distribution channel.	Identifier
Customer_Type_ID	Identifier of the customer classification, deposit type, and meal plan.	Identifier
Reservation_ID	Identifier of the reservation status.	Identifier
is_canceled	Indicates whether the booking was canceled.	Boolean
lead_time	Number of days between booking creation and arrival date.	Days
arrival_date_year	Year of guest arrival.	Year
arrival_date_month	Month of guest arrival.	Month
arrival_date_week_number	Week number of guest arrival.	Week Number
arrival_date_day_of_month	Day of the month of guest arrival.	Day
stays_in_weekend_nights	Number of weekend nights booked.	Nights
stays_in_week_nights	Number of weekday nights booked.	Nights
adults	Number of adult guests included in the booking.	Persons
children	Number of child guests included in the booking.	Persons
babies	Number of babies included in the booking.	Persons
country	Country of origin of the guest.	Country Code
is_repeated_guest	Indicates whether the guest has booked previously.	Boolean
previous_cancellations	Number of previous canceled bookings by the guest.	Count
previous_bookings_not_canceled	Number of previous completed bookings by the guest.	Count
booking_changes	Number of modifications made to the booking.	Count
agent	Travel agent identifier associated with the booking.	Identifier
company	Company identifier associated with the booking.	Identifier
days_in_waiting_list	Number of days the booking remained on a waiting list.	Days
adr	Average Daily Rate charged for the booking.	Currency
required_car_parking_spaces	Number of parking spaces requested by the guest.	Count
total_of_special_requests	Number of special requests submitted by the guest.	Count
reservation_status_date	Date on which the reservation status was recorded.	Date

Hotels table

Column Name	Description	Unit
Hotel_ID	Unique identifier assigned to a hotel.	Identifier
hotel	Name or category of the hotel.	Text

Market Segments table

Column Name	Description	Unit
Customer_Type_ID	Unique identifier assigned to a customer type record.	Identifier
customer_type	Classification of the customer booking behavior.	Category
deposit_type	Deposit policy applied to the booking.	Category
meal	Meal package selected for the booking.	Category

Customer Types table

Column Name	Description	Unit
Market_Segment_ID	Unique identifier assigned to a market segment record.	Identifier
market_segment	Market segment through which the booking was acquired.	Category
distribution_channel	Distribution channel used to make the booking.	Category

Reservation Status table

Column Name	Description	Unit
Reservation_ID	Unique identifier assigned to a reservation status.	Identifier
reservation_status	Final status of the reservation.	Category

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Data Preparation

Data Cleaning and Transformations

Steps performed in Python:

Column Name	Original Type	Updated Type	Action Taken
children	float64	int64	Filled missing values with 0
agent	float64	Int64	Filled missing values with 0 (No Agent)
company	float64	Int64	Filled missing values with 0 (No Company)
country	object	object	Missing values replaced with 'Unknown'
reservation_status_date	object	datetime	Converted to date format
is_canceled	int64	Boolean	Converted 0/1 to Boolean values
is_repeated_guest	int64	Boolean	Converted 0/1 to Boolean values
arrival_date	Derived	datetime	Created from year, month, and day of month fields
total_nights	Derived	int64	Total stay duration created from Weekend Nights + Week Nights
total_price	Derived	float64	Estimated booking value created from ADR × Total Nights

Additional transformations performed in PowerBI :

- Lead Time Categories created for segmentation analysis
- DAX measures created for Average ADR, Average Stay Length, Completed Booking Count, Cancellation Rate, Occupancy Rate, Total Bookings and Total Revenue.

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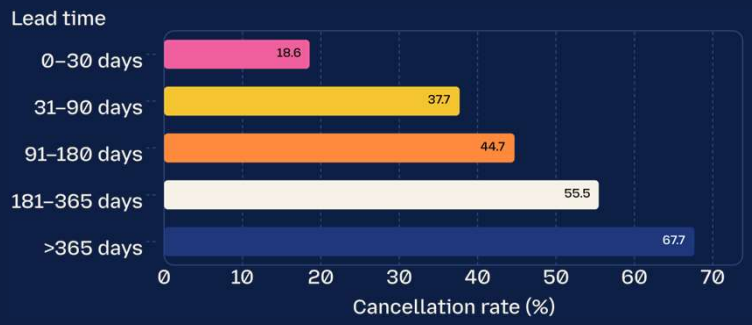
1 - Factors Influencing Booking Cancellations

- Cancellation risk rises with booking lead time

67.66%

cancellation rate for bookings made over 365 days in advance

For 0–30 days, the cancellation rate is 18.56%.



- Deposit and customer mix reveal different kinds of risk

99.36% — **Non-Refund** cancellation rate within the deposit category.

40.75% — **Transient** cancellations highest rate among all customer segments.

10.23% — **Group** cancellations lowest cancellations rate among all customer segments.

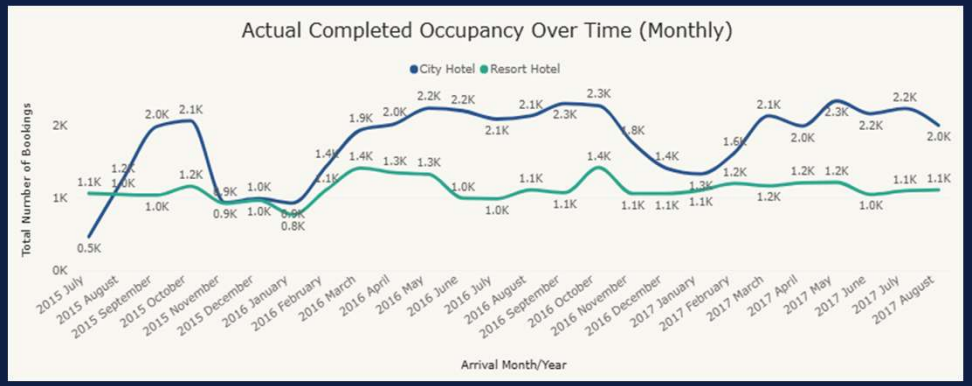
Cancellation Rate by Customer Type & Deposit Type

customer_type	No Deposit	Non Refund	Refundable	Total
Contract	20.28%	100.00%	100.00%	30.96%
Group	10.37%			10.23%
Transient	30.76%	100.00%	100.00%	40.75%
Transient-Party	22.36%	91.80%	10.61%	25.43%
Total	28.38%	99.36%	22.22%	37.04%

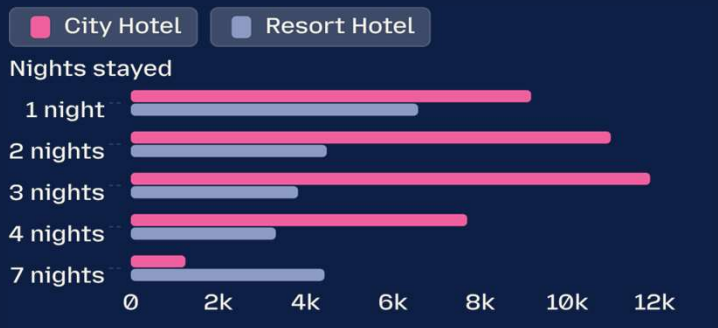
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2 – Occupancy Pattern Analysis

- City Hotel consistently achieved higher occupancy



- Short stays dominate; resort guests stay longer



Completed bookings by stay length

Peak City Hotel

- 2,339 bookings (May 2017)

Peak Resort Hotel

- 1,417 bookings (Oct 2016)

Highest Occupancy Segment

- City Hotel

3.5× more seven-night Resort bookings

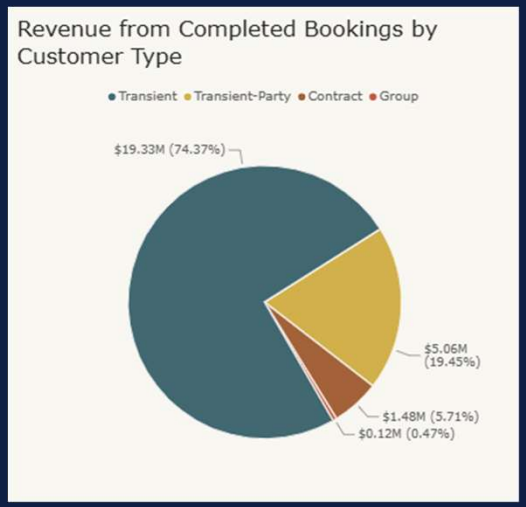
Resort: 4,435; City: 1,251. City peaks at 11,895 bookings for 3 nights.

City bookings cluster in 1–4 nights; Resort has a larger relative share of longer stays.

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3 – Customer Segment Evaluation

- Transient customers generate the highest revenue but carry the greatest cancellation risk





Cancellation rates: Transient 40.75%; Contract 30.96%; Transient-Party 25.43%; Group 10.23%.

4× more revenue

Transient

- Highest revenue ($19.33M)
- Highest ADR ($104.78)
- Highest cancellation rate (40.75%)

Transient-Party

- Second-largest revenue ($5.06M)
- Moderate cancellation rate (25.43%)

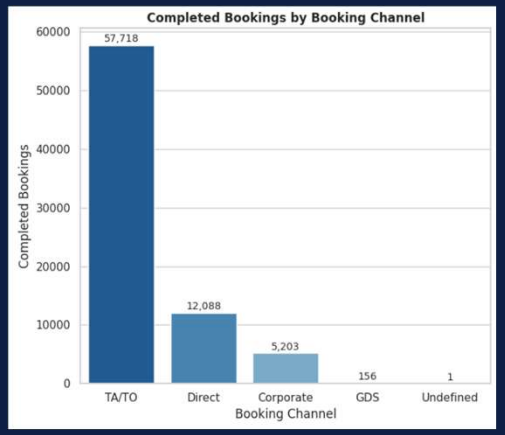
Group

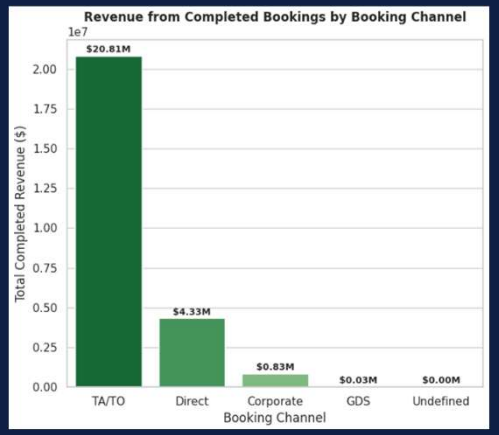
- Lowest revenue ($0.12M)
- Lowest cancellation rate (10.23%)

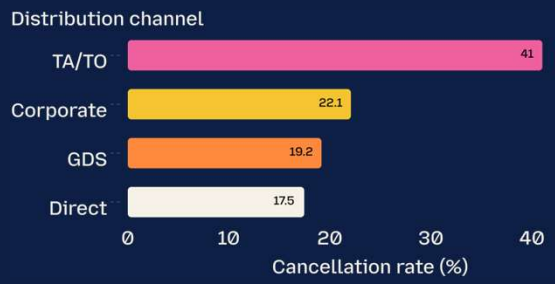
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4 - Distribution Channel Performance

- TA/TO drives scale while Direct delivers the most reliable bookings







TA/TO

- Highest completed bookings (57,718)
- Highest realized revenue ($20.81M)
- Highest cancellation rate (41.03%)

Direct

- Lowest cancellation rate (17.46%)
- Strong revenue ($4.33M)
- Most reliable source of completed stays

Corporate

- Moderate volume and revenue
- More stable than TA/TO

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5 – Reservation Outcome Analysis

- Completed stays show more guest engagement



Average lead time: Canceled 147.3 days; Check-Out 79.98; No-Show 57.18.

Bookings: 43,017 Canceled; 75,166 Check-Out; 1,207 No-Show.

0.71 vs 0.32

Average special requests: Check-Out vs Canceled

Check-Out

- 75,166 bookings, the majority outcome, with the highest average special requests at 0.71.

Cancelled

- Average lead time of 147.3 days, nearly twice completed stays at 79.9 days.

No-Show

- 1,207 bookings, with shorter lead times and moderate request levels.

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Recommended Strategy

- Prioritize actions that improve reliability without suppressing high-value demand.

Priority	Initiative	Target	Action	KPI
1	Risk-tiered booking controls	Long lead time + Transient + OTA/TA/TO	Deposit or guarantee rules, reminder cadence, reconfirmation	Cancellation rate by risk tier
2	Direct conversion program	Repeat and engaged guests	Member-only value, post-stay offer, direct rebooking	Direct share and direct cancellation rate
3	Seasonal occupancy playbook	May and August peaks; hotel-type differences	Staffing, inventory, minimum-stay and package rules	Occupancy, ADR and stay length
4	Engagement-to-arrival workflow	Guests with few or no requests	Pre-arrival preference capture and service prompts	Request rate and stay completion

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Hotel Booking Performance Dashboard

Data from July 2015 to Aug 2017

$26M	75K	62.96%	37.04%
Total Completed Revenue	Total Completed Bookings	Occupancy Rate	Cancellation Rate

Actual Completed Occupancy Over Time (Monthly)



Revenue Contribution by Distribution Channel



Customer Type: All

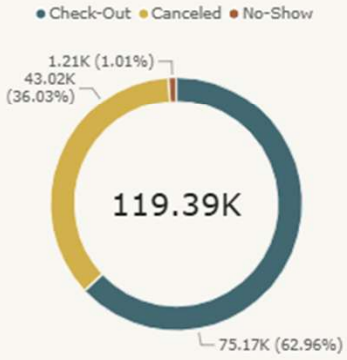
Hotel: All

Market Segment: All

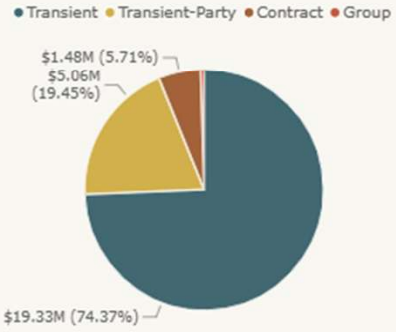
Reservation Status: All

Arrival Date: 1/7/2015 – 31/8/2017

Reservation Outcomes



Revenue from Completed Bookings by Customer Type



Cancellation Rate by Customer Type & Deposit Type

customer_type	No Deposit	Non Refund	Refundable	Total
Contract	20.28%	100.00%	100.00%	30.96%
Group	10.37%			10.23%
Transient	30.76%	100.00%	100.00%	40.75%
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Total	28.38%	99.36%	22.22%	37.04%

Cancellation Rate by Lead Time



The management decision

Grow the demand that stays.

Use risk-tiered controls for high-value volatile bookings, shift repeatable demand toward Direct, and operationalize seasonal occupancy patterns.

Protect
high-revenue demand

Convert
engaged guests to Direct

Plan
by season and hotel type

Measure
completion, not bookings alone

Q&A

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